# HR ANALYTICS

# Employee Attrition Analysis & Prediction

PROJECT REPORT + MODEL EVALUATION

| | |
|---|---|
| **Project focus** | Explore employee attrition and evaluate classification models. |
| **Tools** | Python, Pandas, scikit-learn, SHAP, Power BI |
| **Scripts** | hr_analytics.py; hr_analytics_model.py; SHAP analysis script |
| **Dashboard** | Reported as completed by the project owner. |

**Important: the numerical model scores are left blank until actual saved outputs are supplied. This avoids inventing results.**

Student: Ranaprathap Chittimelli

Submission date: _____

# 1. Abstract

Employee attrition occurs when employees leave an organization. High attrition can increase recruitment and training costs and disrupt teams. This project combines exploratory data analysis (EDA), Power BI visualization, classification models, and SHAP explainability to study attrition patterns and demonstrate a repeatable prediction workflow. The work includes data preparation, exploratory charts, Logistic Regression and Decision Tree models, evaluation of predictive performance, and interpretation of model behaviour using SHAP.

Final conclusions should be based on the actual dashboard, evaluation metrics, and SHAP plots. A prediction indicates a statistical pattern, not proof that a feature causes an employee to resign.

# 2. Problem Statement

HR teams need a structured way to review employee attrition patterns and identify records that may need further investigation. Reviewing a large dataset manually can be time-consuming. This project uses descriptive analytics and machine learning to summarize patterns and demonstrate a repeatable prediction workflow.

# 3. Objectives

• Inspect and prepare employee data.

• Explore attrition patterns across employee characteristics.

• Build and compare Logistic Regression and Decision Tree classifiers.

• Evaluate models using accuracy, precision, recall, F1-score, and confusion matrices.

• Use SHAP plots to explain model predictions.

• Present summaries in Power BI and develop evidence-based HR recommendations.

# 4. Dataset Description

The project has been described as using the IBM HR Analytics Employee Attrition dataset. The commonly distributed version contains 1,470 employee records and 35 columns. Confirm these counts against the exact CSV/XLSX file used for your analysis before submission.

| Item | Description | Check before submission |
|---|---|---|
| Target | Attrition (Yes / No) | Confirm values in the source file |
| Features | Employee, job, compensation, satisfaction, tenure fields | List final features used by model |
| Rows / columns | Common public version: 1,470 / 35 | Verify dataset shape in Python |

# 5. Tools and Technologies

| Tool | Purpose |
|---|---|
| Python / Pandas | Data loading, preparation, transformation, and analysis. |
| Matplotlib / Seaborn | EDA charts and visual analysis. |
| scikit-learn | Train/test split, classifiers, and evaluation metrics. |
| SHAP | Explaining feature contributions to predictions. |
| Power BI / DAX | Interactive dashboard, KPI measures, and visual comparisons. |

# 6. Exploratory Data Analysis (EDA)

EDA helps understand the data before modelling. Relevant checks include missing values, duplicate records, attrition class distribution, attrition by department and job role, income bands, overtime, years at company, satisfaction, and promotion-related fields where available. Use the generated charts to confirm which analyses were actually performed.

Record only patterns observed in the real charts. Do not describe an association as a cause. Insert the strongest EDA plots if visual evidence is required.

# 7. Power BI Dashboard

The project owner reports that all dashboard visuals are complete. The final report should include screenshots of the actual dashboard and a short explanation of each visual. Verify KPI values against the source dataset and test that slicers filter the intended visuals.

| Dashboard item | What to document |
|---|---|
| KPI cards | Record displayed totals and validate them against the source dataset. |
| Department / role charts | Describe the highest observed counts only after checking the visual. |
| Salary / tenure visuals | Explain observed patterns; note that band thresholds are analytical choices. |
| Slicers | Confirm the selected filters affect relevant visuals. |

## Dashboard screenshot placeholder

[Insert a screenshot of your completed Power BI dashboard in the final editable report.]

# 8. Machine Learning Methodology

The project compares Logistic Regression and Decision Tree classifiers. A reproducible workflow defines the target, separates inputs from the target, splits data into training and test sets, preprocesses categorical and numerical features without test-data leakage, trains both models, and evaluates them on the same held-out test set.

## Evaluation metrics

Accuracy measures the overall fraction of correct predictions. Precision measures how many predicted attrition cases were actual attrition cases. Recall measures how many actual attrition cases were detected. F1-score is the harmonic mean of precision and recall. For attrition, pay particular attention to recall and F1-score for the 'Yes' class, not accuracy alone.

## Confusion matrix

A confusion matrix counts true negatives, false positives, false negatives, and true positives. Record the class order used by your code because interpretation depends on which class is treated as positive.

# 9. Model Evaluation Results

**The numerical scores are intentionally not fabricated. Copy them from outputs/model_comparison.csv or recalculate them from the saved test labels and predictions.**

| Metric | Logistic Regression | Decision Tree |
|---|---|---|
| Accuracy | _____ | _____ |
| Precision (Attrition = Yes) | _____ | _____ |
| Recall (Attrition = Yes) | _____ | _____ |
| F1-score (Attrition = Yes) | _____ | _____ |
| Confusion matrix file | _____ | _____ |

## How to interpret the comparison

After filling the table, compare the models using recall and F1-score for Attrition = Yes, precision, overall accuracy, and the confusion matrices. Consider the business cost of false negatives. Do not choose a model on accuracy alone or claim reliability without checking the test procedure and class balance.

## Python snippet to calculate metrics

Use this if your existing script already has y_test, y_pred_lr, and y_pred_dt. Ensure the positive label matches the encoding used in your code.

```
from sklearn.metrics import classification_report, confusion_matrix

print("LOGISTIC REGRESSION")
print(classification_report(y_test, y_pred_lr, zero_division=0))
print(confusion_matrix(y_test, y_pred_lr))

print("DECISION TREE")
print(classification_report(y_test, y_pred_dt, zero_division=0))
print(confusion_matrix(y_test, y_pred_dt))
```

# 10. SHAP Explainability

SHAP estimates how features contribute to a model prediction relative to a baseline. Use the actual SHAP summary and feature-importance plots to identify features with the strongest overall influence. A global importance plot shows the magnitude of contribution; a summary beeswarm plot can also show whether values tend to push predictions higher or lower, depending on the model output and class being explained.

Insert the generated plots and write 2–4 observations based on visible feature names and directions. SHAP explains model behaviour; it does not establish causality.

[Insert SHAP summary plot here.]

[Insert SHAP feature-importance plot here.]

# 11. Findings and HR Recommendations

Complete this section after reviewing the real dashboard and SHAP outputs. Use the following evidence-led structure:

| Field | Complete with project evidence |
|---|---|
| Finding 1 | The dashboard indicates that _____. |
| Evidence | This is supported by the visual / metric _____. |
| Finding 2 | The SHAP results show that _____ influences predictions. |
| Action to investigate | HR could review _____ and evaluate whether an intervention helps. |

### Possible actions to consider (not conclusions unless supported by evidence)

Review overtime workload and employee feedback; examine career development and promotion processes; investigate job satisfaction and manager support; review compensation bands for equity; and monitor attrition trends by department and role. Validate proposed interventions with HR context and follow-up measurement.

# 12. Limitations and Ethical Considerations

The dataset may not represent every organization or current workforce. Historical patterns may not generalize to other employers. Scores depend on splitting, preprocessing, feature selection, class imbalance, and threshold choice. Some fields may act as proxies for sensitive or protected characteristics. Predictions should not be the sole basis for employment decisions or individual treatment. Use them only as decision support, protect privacy, and evaluate fairness and potential harm.

# 13. Conclusion

This project demonstrates a workflow for analysing employee attrition through EDA, Power BI visualization, supervised classification, and SHAP explainability. The final conclusions should report observed dashboard patterns, verified model metrics, and actual SHAP findings. Possible extensions include cross-validation, threshold tuning, fairness checks, and validation on additional data.

# 14. Final Submission Checklist

- ■ Verify dataset row/column counts and target distribution.
- ■ Add actual model metrics and confusion matrices.
- ■ Insert Power BI dashboard screenshot(s).
- ■ Insert SHAP plots and evidence-based interpretations.
- ■ Complete findings and recommendations from actual outputs.
- ■ Include scripts and README in GitHub.
- ■ Remove placeholders and proofread before submission.

# Appendix: Project files

Keep hr_analytics.py, hr_analytics_model.py, the SHAP analysis script, the Power BI .pbix file, model_comparison.csv, confusion-matrix images, SHAP plots, requirements.txt, README.md, and the final PDF. Include the dataset only where permitted and avoid private employee information.

**Report status: draft template — fill the evaluation values and insert actual evidence before formal submission.**